

Exercise 1

Consider the system $x(k+1) = Ax(k) + Bu(k)$, $x \in \mathcal{X}$, $u \in \mathcal{U}$ with

$$\mathcal{X} = \{x \in \mathbb{R}^n | H_x x \leq b_x\}$$

$$\mathcal{U} = \{u \in \mathbb{R}^m | H_u u \leq b_u\}$$

and stage cost $l(x, u) = x^\top Q x + u^\top R u$ with $Q \in \mathbb{R}^{n \times n}$, $R \in \mathbb{R}^{m \times m}$ positive definite. Find a state-feedback $\kappa_f(x) = Kx$, $K \in \mathbb{R}^{m \times n}$, a corresponding terminal cost $V_f(x) = x^\top P x$, $P \in \mathbb{R}^{n \times n}$, and a terminal set $\mathcal{X}_f \neq \{0\}$ such that the following conditions are satisfied for all $x(k) \in \mathcal{X}_f$:

1. Terminal set is **invariant** under the local control law $\kappa_f(x(k))$:

$$x(k+1) = Ax(k) + B\kappa_f(x(k)) \in \mathcal{X}_f.$$

2. All state and input **constraints are satisfied** in $\mathcal{X}_f$:

$$\mathcal{X}_f \subseteq \mathcal{X}, \quad \kappa_f(x(k)) \in \mathcal{U}.$$

3. Terminal cost is a continuous **Lyapunov function** in the terminal set $\mathcal{X}_f$ and satisfies:

$$V_f(x(k+1)) - V_f(x(k)) \leq -l(x(k), \kappa_f(x(k))).$$

Solution:

Approach:

- Use LQR control law as terminal controller $\kappa_f(x) = F_\infty x$, i.e. $K = F_\infty$ and LQR infinite horizon cost as terminal cost $V_f(x) = x^\top P_\infty x$, using Q and R given, see Lec. 2, Slides 40-41.
- Yields $V_f((A + BF_\infty)x) - V_f(x) \leq -l(x, F_\infty x)$ (Lec. 6, Slides 48-49)
- Choose $\mathcal{X}_f(\beta) = \{x | x^\top P_\infty x \leq \beta\}$ as terminal set.
- Invariant set for the unconstrained system for any β (Lec. 5, Slide 70)

What about constraint satisfaction?

- Find maximum β such that state and input constraints are satisfied: $\mathcal{X}_f \subseteq \mathcal{X}$, $F_\infty x \in \mathcal{U}$, for all $x \in \mathcal{X}_f$.

$$\begin{aligned} \max \quad & \beta \\ \text{s.t.} \quad & \mathcal{X}_f(\beta) \subseteq \mathcal{X} = \{x \in \mathbb{R}^n | H_x x \leq b_x\} \\ & \mathcal{X}_f(\beta) \subseteq \mathcal{X}_u = \{x \in \mathbb{R}^n | H_u F_\infty x \leq b_u\} \end{aligned}$$

- Recall Lec. 5, Slide 71-72:

Support function for single constraint $F_i x \leq f_i$:

$$h_{\mathcal{X}_f(\beta)}(F_i^\top) = \max_x F_i x \quad \text{s.t. } x^\top P_\infty x \leq \beta$$

For ellipsoidal sets:

$$h_{\mathcal{X}_f(\beta)}(F_i^\top) = \|P_\infty^{-1/2} F_i^\top\| \sqrt{\beta}$$

- Scaling such that $\mathcal{X}_f(\beta)$ is within state constraints:

$$\begin{aligned} \max \quad & \beta_x \\ \text{s.t.} \quad & h_{\mathcal{X}_f}(H_{x,i}^\top) \leq b_{x,i}, \text{ for all } i \in \{1, \dots, n_{\text{constr},x}\}, \end{aligned}$$

$$\text{solution } \beta_x^* = \min_{i \in \{1, \dots, n_{\text{constr},x}\}} \frac{b_{x,i}^2}{H_{x,i} P_\infty^{-1} H_{x,i}^\top}$$

- Scaling such that $\mathcal{X}_f(\beta)$ is within input constraints: Recall we mapped the input constraints to state constraints above, i.e. $H_{u,i} u \leq b_{u,i} \rightarrow H_{u,i} F_\infty x \leq b_{u,i}$, and require input satisfaction via

$$\begin{aligned} \max \quad & \beta_u \\ \text{s.t.} \quad & h_{\mathcal{X}_f}(F_\infty^\top H_{u,i}^\top) \leq b_{u,i}, \text{ for all } i \in \{1, \dots, n_{\text{constr},u}\}, \end{aligned}$$

$$\text{solution } \beta_u = \min_{i \in \{1, \dots, n_{\text{constr},u}\}} \frac{b_{u,i}^2}{H_{u,i} F_\infty P_\infty^{-1} F_\infty^\top H_{u,i}^\top}$$

- Take minimum, i.e. $\beta = \min\{\beta_x, \beta_u\}$

Exercise 2

Consider the system

$$x(k+1) = \begin{bmatrix} 1.1 & 0.1 \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} 0 \\ 0.1 \end{bmatrix} u(k) \quad (1)$$

with initial condition $x(0) = [-0.7, 1.4]^\top$, state constraints $\mathcal{X} = \{x \in \mathbb{R}^2 : \|x\|_\infty \leq 2\}$, input constraints $\mathcal{U} = \{u \in \mathbb{R} : \|u\|_\infty \leq 1\}$.

- Write the constraints $\mathcal{X}$ and $\mathcal{U}$ in polytopic form, i.e., $\mathcal{X} = \{x \in \mathbb{R}^2 | H_x x \leq h_x\}$ and $\mathcal{U} = \{u \in \mathbb{R} | H_u u \leq h_u\}$.
- Define the system dynamics, constraints and initial condition in the function `generate_params`.
- In the class `MPC_DEMO`, implement the following model predictive controller using Yalmip¹

$$\min_{x,u} \sum_{i=0}^{N-1} x_i^\top x_i + u_i^\top u_i \quad (2a)$$

$$\text{s.t. } x_0 = x(k) \quad (2b)$$

$$x_{i+1} = A x_i + B u_i, \quad i = 0, \dots, N-1 \quad (2c)$$

$$H_x x_i \leq h_x, \quad i = 0, \dots, N-1 \quad (2d)$$

$$H_u u_i \leq h_u, \quad i = 0, \dots, N-1 \quad (2e)$$

$$x_N = 0. \quad (2f)$$

¹<https://yalmip.github.io/>, included in the Multi-Parametric Toolbox (MPT): <https://www.mpt3.org/>. See also the installation instructions in the project description of the MPC programming exercise.

Compare and discuss the closed-loop simulation of the controller *with and without* terminal constraint for different horizon lengths $N \in \{30, 15, 11, 10\}$.

Solution:

- i) From the definition of the infinity norm we have $-2 \leq x_1 \leq 2$, $-2 \leq x_2 \leq 2$, and $-1 \leq u \leq 1$.
We have

$$\begin{aligned}x_1 &\leq 2 \\ -x_1 &\leq 2 \\ x_2 &\leq 2 \\ -x_2 &\leq 2 \\ u &\leq 1 \\ -u &\leq 1\end{aligned}$$

It follows that

$$H_x = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \\ 0 & -1 \end{bmatrix}, h_x = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}, H_u = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, h_u = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

- ii) See provided MATLAB code.
iii) See provided MATLAB code.